

Development and validation of a pulse wave velocity monitoring device

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ABSTRACT

Cardiovascular diseases are known to be the main causes of death worldwide. Although the number of deaths due to cardiovascular diseases decreased from 2000 to 2010, there was an increase in cardiovascular disease-related deaths from 2010 to 2020. Comprising heart attacks, hypertension, diabetes, atherosclerosis, and other minor diseases, this group is associated with changes in arterial stiffness, which refers to the artery's ability to expand and contract as pressure varies during the cardiac cycle. Recent guidelines from the European Society of Cardiology (2023) regarding the treatment of arterial hypertension recommend using pulse wave velocity (PWV) measurement. This method is the most reliable way to assess arterial stiffness. PWV measurement is considered the gold standard for monitoring arterial stiffness. Therefore, the development of new sensors capable of measuring PWV has been a research focus in recent years, as commercial systems are expensive and difficult to operate. In this context, this work presents the design of two arterial pulse sensors for detecting PWV. After characterization, the system showed a root mean square deviation (RMSD) between the detected and the reference waves of 0.0110 and 0.0196 for each probe, allowing to extract value information from the signals in the time domain. The results of PWV were 12.17, 13.35, and 14.67 m/s, for three different pressure values i.e. 70, 80, and 90 mmHg in the testbench, showing the feasibility of the proposed approach to measure PWV. As expected, an increase in pressure corresponds to an increase in pulse wave velocity.

Keywords: Optical fiber sensors, polymer optical fiber, wave velocity monitoring, cardiovascular diseases

1. INTRODUCTION

According to the research Cardiovascular Statistics - Brazil 2023, cardiovascular diseases contributed to 9.89 million deaths in the world in 2021, with 382.507 deaths in Brazil. Among men, coronary artery disease (CAD) was ranked as the second most prevalent cause of death among the top ten causes, right behind COVID-19. Among women, CAD ranked as the third leading cause of death, following COVID-19 and cerebrovascular disease, with corresponding mortality rates of 6.9% for men and 6.0% for women [1]. Therefore, monitoring heart risk factors is essential for preventing heart disease. In this context, the American Heart Association recommended 7 aspects that should be assessed in an individual for quality cardiovascular health, divided into 3 behavioral aspects (diet quality, physical activity, and smoking) and 4 key factors (presence of blood cholesterol, body mass index (BMI), blood pressure, and blood glucose levels) [2]. In addition, arterial stiffness is another factor proven to be a strong predictor of cardiovascular events such as coronary artery disease [3], heart attack [4], hypertension [5], and diabetes [6]. Arterial stiffness results from changes in the structure of the arterial wall, which loses elasticity with age and consequently increases the pulse wave propagation. The arterial wall stiffness increment is associated with the balance of two proteins: collagen and elastin [7] and is also affected by conditions such as being overweight, smoking, and alcoholism.

To assess the arterial stiffness, the pulse wave velocity through the artery is measured. This method is particularly valuable in assessing arterial stiffness, which is commonly equated to pulse wave velocity (PWV), the speed at which the arterial pulse propagates along the arterial wall. In clinical practice, PWV is typically calculated as the ratio of the distance between two measurement sites to the time it takes for the pulse to travel between them [8]. Nowadays, non-invasive commercial devices are used for clinical diagnostics utilizing tonometry, piezoelectrics, arteriography, photoplethysmography, ultrasound, and magnetic resonance imaging [9]. However, these devices have some disadvantages related to the cost and/or use of electronic sensors in contact with the patient. Arathy *et al.* [10] used accelerometers applied to two points on the carotid artery to acquire cardiac pulse signals, presenting the design and validation of the proposed system. Nguyen *et al.* [11] demonstrated the development of a sensor based on a

piezoresistive structure embedded in an air chamber applied to the wrist and the tip of the index finger, showing greater sensitivity compared to sensors of the same type embedded in other materials such as gel and liquid. Nabeel *et al.* [12] developed a device using photoplethysmography. The sensors were applied to two adjacent points on the carotid artery, 28 cm apart, allowing the acquisition of the cardiac pulse at each location to calculate pulse wave velocity. This study was conducted with 10 healthy volunteers to demonstrate the feasibility of the proposed system.

An alternative technology to overcome the issues of the aforementioned approaches is the optical fiber sensors (OFS), which present advantages such as immunity to electromagnetic interference, inert to chemical and biological agents, biocompatibility, small size and low weight, and remote monitoring, among others [13]. The development of OFS has grown across various fields, including industry, with applications like temperature [14], pressure [15], level [16], and vibration sensors [17] and medicine, where new devices have emerged to aid in biomechanical rehabilitation [18] and disease diagnosis [19]. Among OFS, fiber Bragg grating (FBGs) are broadly used, however, they suffer from bulk and high-cost interrogation systems [20], [21]. Intensity-based sensors made from polymeric optical fibers (POF) have been used due to their simplicity and low-cost interrogation systems. To monitor the cardiac pulse and evaluate arterial PWV, Leitão *et al.* [22] developed a sensor using a plastic box where an FBG was placed to detect the movements of a sphere caused by the distension of the carotid artery. Leitão *et al.* also studied arterial pulse detection through the variation of optical power returning to the optical fiber via a reflective element in contact with the skin. The sensor characterization experiments and tests conducted on individuals demonstrated the effectiveness of the proposed system, and the data analysis indicated its potential as an alternative to the tonometer for arterial health assessment detecting the pulse and monitoring patients' vital signs [23]. Haseda *et al.* [24] presented a sensor based on plastic optical fiber Bragg gratings (POF-FBG) for measuring pulse wave signals and blood pressure. When the fiber is subjected to pressure, the FBG experiences changes in its periodicity, resulting in a shift in the reflected Bragg wavelength. This shift is proportional to the strain or pressure applied to the fiber. In their setup, the fiber was placed directly on the skin, enabling non-invasive measurement of physiological signals.

In this context, this work presents a POF-based sensor to monitor the pulse wave propagation at two distinct points on the body. The sensor was developed using 3D printing techniques and characterized in an experimental setup that allows to set both the fiber tip distance and the pulse wave propagation. Chapter 2 presents the sensor's development, the experimental setup, and the experimental validation. Chapter 3 presents the results and, finally, the conclusion and future works are drawn in Chapter 4.

2. MATERIALS AND METHODS

2.1 Optical Fiber Sensor

Figure 1 presents the proposed sensor [25]. The sensor's operating principle is based on light intensity variation. The light into the cavity is backscattered to the optical fiber by the probe tip (Figure 1a). Mechanical motion on the probe tip (artery's pulse) causes variations in the internal cavity length which modulates the light. The spring and the cap guarantee the probe tip returns to its initial position when pressure is removed. To send the light to the internal cavity and to couple the backscattered light, a 1x2, 50/50 power splitter (IF-562, Industrial Fiber Optics, USA) was used. The light source is a 660 nm light-emitting diode (LED, IF-E97, Industrial Fiber Optics, USA), and the photodetector (PD) is a phototransistor (IF-D92, Industrial Fiber Optics, USA).

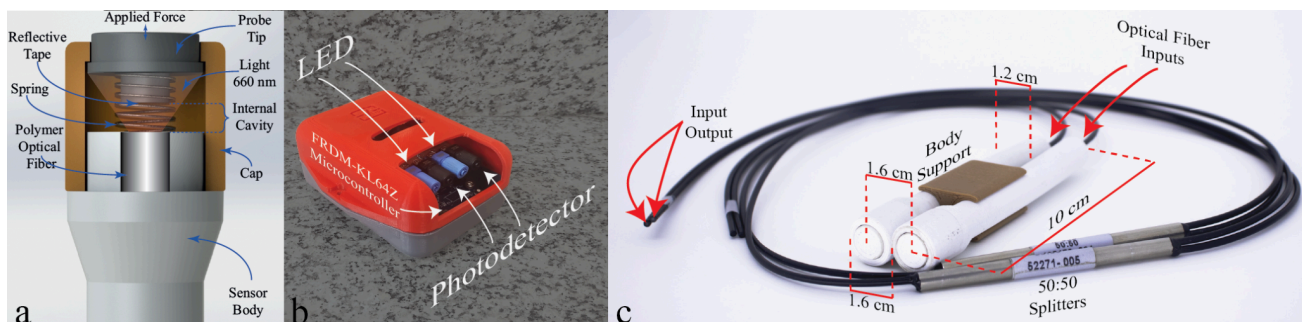


Figure 1. (a) Cross-sectional view of the sensor's internal mechanism. (b) Interrogation system. (b) Sensor assembly.

Figure 1c depicts the proposed sensor. The minimum distance between probes is 1.6 cm which is limited by the body support. The sensor dimensions are 10 cm in length, 1.2 cm in diameter in the body, and 1.6 cm in diameter in the tip. Those dimensions allow an ergonomic grip and make putting the sensor in the carotid artery easy. As can be seen, it is possible to remove each probe from the body support. This allows more flexibility putting each probe tip at different distances freely. To activate the LEDs and read the optical signals with the PDs, an electronic system based on a microcontroller ARM Cortex-M4, with an on-chip 16-bit resolution analog to digital converter (ADC) was used (FRDM-K64F, NXP, Netherlands). The 3D-printed box with electronics inside is shown in Figure 1b.

2.2 Experimental Setup

The probes were tested in a bench setup to evaluate the captured wave behavior against the modulated signal. Figure 2 illustrates the system used to generate mechanical pulses in the sensors, comprising a piezoelectric actuator (APFH 720, ThorLabs), a controller (KPZ101, ThorLabs), and the developed POF sensors. Each probe was positioned in a 3D-printed structure anchored to an optical breadboard allowing the actuator's amplitude variations to induce movement at its tip. Carotid distension waves were generated using a data acquisition device (DAQ, USB-6211, National Instruments) and transmitted to the controller, where they were converted into the piezoelectric voltage range. The signals were then collected via the controller's monitoring output and processed by the DAQ. Sensor behavior was monitored through the proposed optoelectronic circuit.

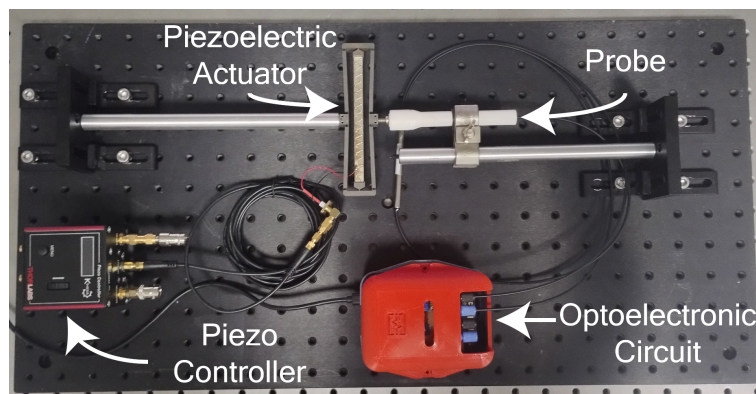


Figure 2. Experimental setup to characterize the sensor response to mechanical deformations.

After characterization, the sensor's performance was analyzed through a test bench that simulated pulse waves (Figure 3). The experimental setup consists of a piezoelectric actuator (P287.70, Physic Instrumente) controlled by a high-voltage piezo driver (E-501.00X, PI-USA). A mixture of water and glycerine (50% each) simulates the blood density. This mixture is housed in a silicon tube of 100 cm in length and 10 mm in diameter. To control the internal pressure system, a syringe and a reference sensor (XGZP6857100KPGPN) were used. The electrical signals were generated through an arbitrary waveform generator (AWG, Agilent 33220A) which supplies the amplifier with the desired signal i.e. short pulses with variable amplitude and period. A graphical user interface (GUI) in LabVIEW software measures the current pressure values from the reference sensor and the signal that feeds the amplifier via a DAQ (USB-6009, National Instruments). The acquisition GUI was developed in Python which allows saving the data from the POF sensors through the interrogation system. The tested distances ranged from 10 cm to 60 cm, in increments of 10 cm, and for each distance evaluated, an average of eight readings was taken. Each reading was conducted over 15 seconds. The beats were generated varying from 0.5 to 4.0 Hz, in increments of 0.5 Hz, corresponding to a range of 30 to 240 BPM for each distance analyzed. The experiment was repeated for pressure values of 70, 80, and 90 mmHg. The probes were anchored to a movable pole in order to change the distance between them.

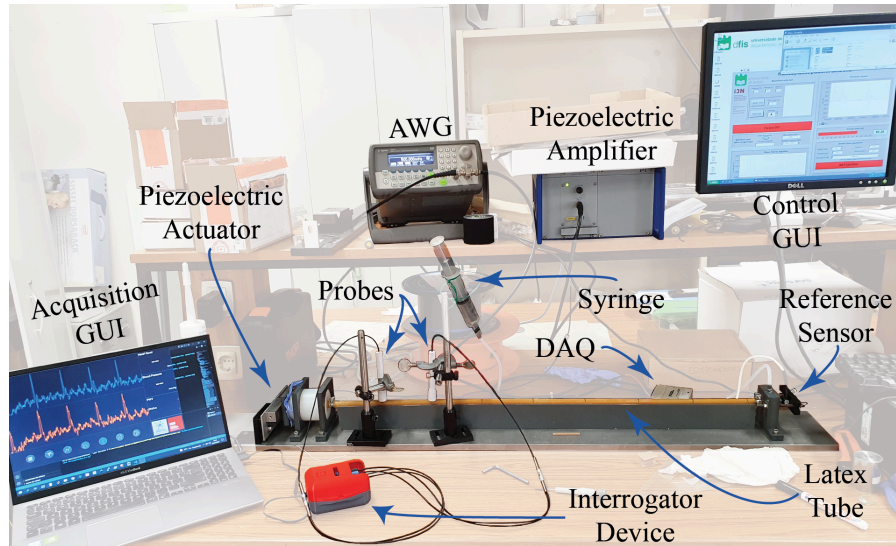


Figure 3. Experimental setup to characterize the sensor response to mechanical deformations.

3. RESULTS AND DISCUSSION

The proposed POF sensor demonstrated promising results during its experimental validation. The setup evaluated the system's ability to detect pulse wave shape and measure PWV across different scenarios in controlled laboratory conditions. The first bench setup allowed for a detailed evaluation of the waveforms captured by the probes in relation to the modulated signal generated by the piezoelectric actuator. Each probe was positioned to detect variations in the actuator's amplitude, inducing movement at the probe tip. The results showed that the root mean square deviation (RMSD) between the detected wave and the reference wave was 0.0110 for the first probe and 0.0196 for the second probe. The analysis of the pulse images revealed that the signal's peak remained unaffected as shown in Figure 4, enabling precise detection of the moment the pulse passed through the sensors. This test demonstrated the feasibility of calculating pulse wave velocity, showcasing the sensor's potential for applications in arterial stiffness assessment.

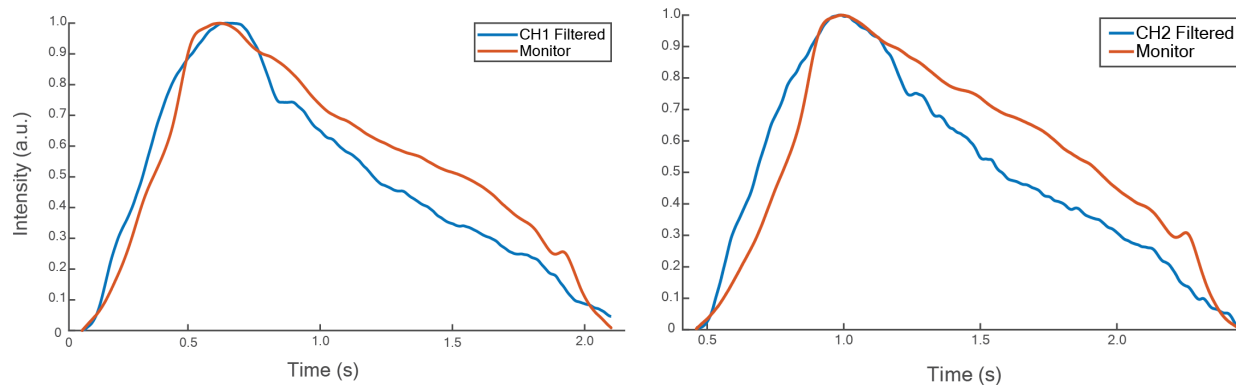


Figure 4. Response of each probe compared to the reference signal.

The proposed sensors were further tested using a simulated pulse wave system to evaluate their performance under controlled conditions. The results demonstrated a strong linear relationship between the measured delay and the distance between sensors for all tested conditions. Specifically, the determination coefficients (r^2) were 0.99, 0.97, and 0.95 for pressure values of 70, 80, and 90 mmHg, respectively. For the three pressure levels tested the calculated PWV values were 12.17, 13.35, and 14.67 m/s, respectively. It is observed as the pressure increases the PWV too. These results indicate the sensor's ability to detect the pulse propagation delay. This capability is crucial for accurately calculating PWV, a key parameter for assessing arterial stiffness. The findings validate the reliability of the proposed system for clinical applications, particularly in scenarios requiring precise and non-invasive monitoring of cardiovascular health.

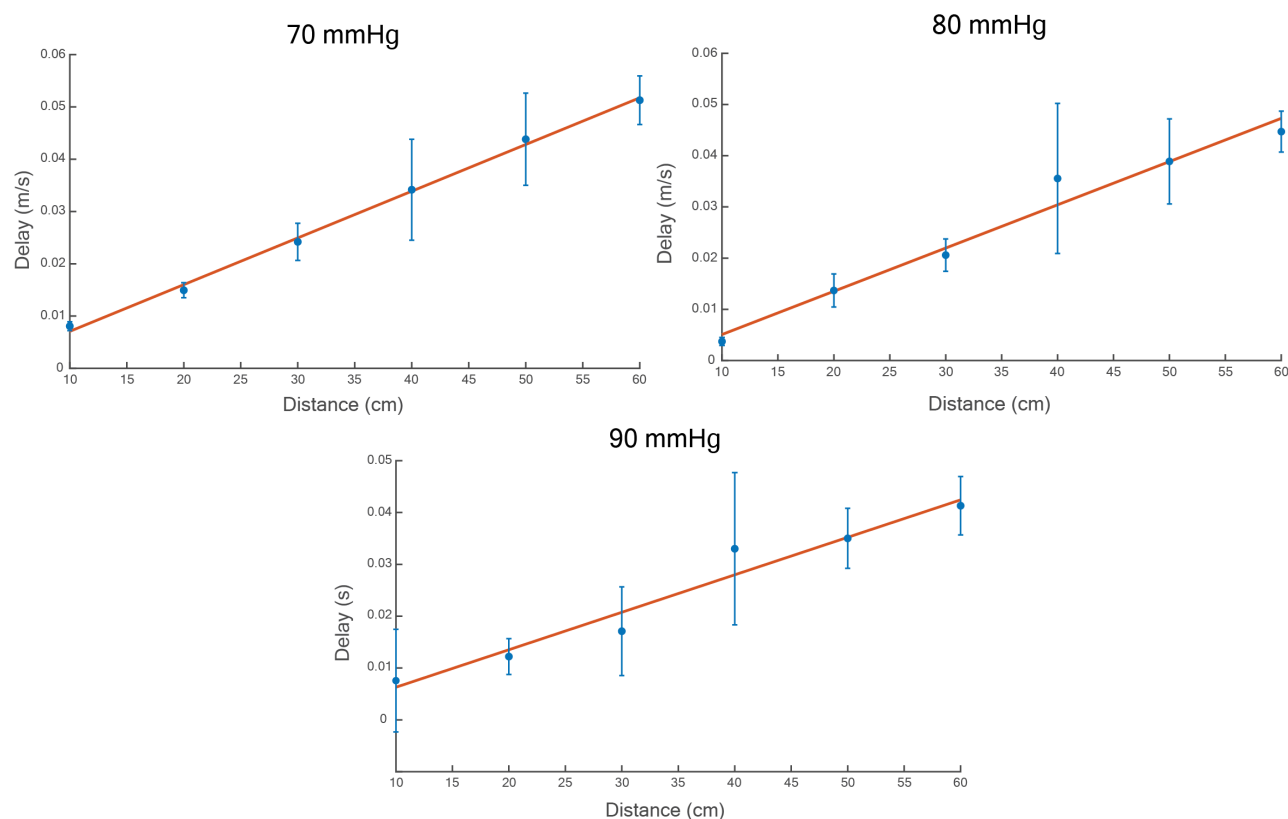


Figure 5. Variation of delay as a function of distance with linear fitting at 70, 80, and 90 mmHg.

4. CONCLUSIONS AND FUTURE WORK

This study presented the development of a cardiac waveform and pulse wave velocity monitoring system using POF-based sensors. The system was validated through laboratory tests and experiments with a healthy individual [25]. The implementation of a graphical interface enabled real-time data processing and visualization, enhancing the system's usability for future applications, especially in scenarios requiring immediate patient evaluation. The proposed device demonstrated an R^2 of 0.99, 0.97, and 0.95 for measurements of 70, 80, and 90 mmHg, respectively. This indicates that the system possesses adequate resolution and data acquisition speed to detect pulse velocity, suggesting its potential applicability in human subjects. For future work, efforts should focus on improving signal processing techniques to estimate blood pressure and conducting clinical trials with a group of participants, including individuals with various cardiovascular conditions, to evaluate the device's performance in real-world scenarios. Comparative studies with existing commercial devices are also recommended to validate the system's clinical applicability. Evaluating the interface design and the system's usability by healthcare professionals will be important for refining its implementation in clinical settings. Furthermore, exploring the possibility of monitoring additional vital signs or physiological parameters could expand the device's utility across different healthcare applications. In summary, this work establishes a solid foundation for an accessible cardiovascular monitoring system based on optical fiber sensors, while offering numerous opportunities for further research and development to enhance its clinical relevance and usability.

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